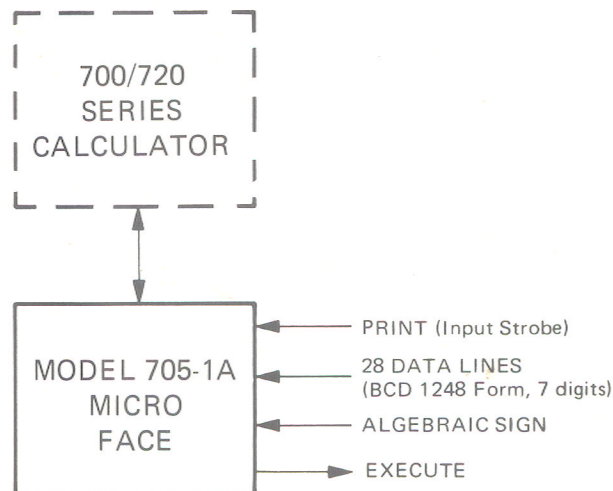
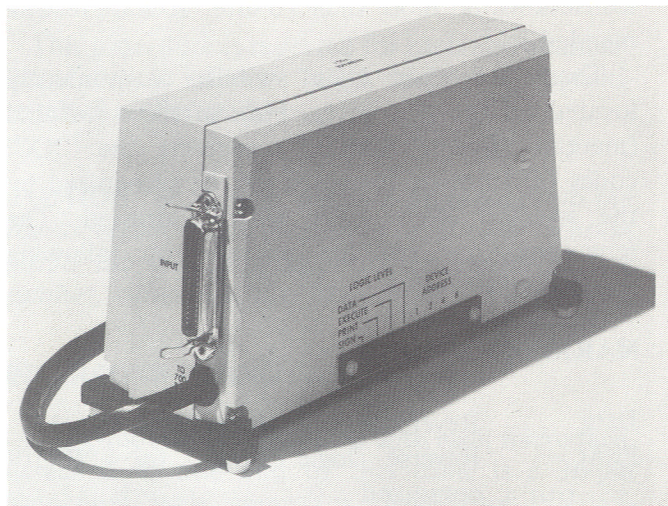


WANG MODEL 705-1A MICRO FACE



GENERAL INFORMATION

The Model 705-1A Micro Face allows a variety of external digital devices (digital voltmeters, panel meters, counters, digital clocks, etc.) to be interfaced to the Model 700 Series Programmable Calculators for many on-line applications. As an input-only interface, the Model 705-1A can easily accept numeric data of up to seven digits plus algebraic sign in parallel using standard BCD 1, 2, 4, 8 format. The Micro Face consists of an input connector, a code converter which converts standard BCD 1, 2, 4, 8 codes into the hexadecimal codes required for the programmable calculator language, a parallel-to-serial converter, and two control signals. Design conveniences of this interface enables adaption to fixed or floating point entry for automatic ranging.

INPUT DATA SIGNAL LEVELS

Input circuits are TTL/DTL compatible. (See Figure 1 for typical input circuit.) Voltage levels for a logic "0" are between 0 to +0.4 volts and voltage levels for a logic "1" are between +2.5 to 5 volts. A convenient slide switch is located on the

rear cover to reverse these logic definitions to the complement bar functions (i.e., 1 = 0 to 0.4 volts and 0 = +2.5 to 5 volts). Input impedance is typically 4K ohms with a 47K ohm resistor in parallel which requires approximately 1.6 milliamp sink (see Figure 1). The pin diagram (Figure 3) shows signal input digit positions, as well as control signals and their associated BCD codes.

CONTROL SIGNALS

The Micro Face allows for one input control signal ("Print" command) and provides one output control signal ("Execute" command).

The "Print" command, which is an input control signal, indicates that information is present on the input lines and is available for transfer. This signal should be connected to pin 18 of the input connector (Figure 3). The "Print" command (input strobe) is normally a d.c. condition of 0 or +5 volts. A convenient slide switch is available to reverse these logic levels. If a pulse is used for the "Print" command, a pulse width of approximately 5 microseconds is sufficient to be recognized by the Micro Face.